Space-State Evolution Theory

and Its View of Time

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Abstract. This paper proposes *Space-State Evolution Theory*, which takes space and its physical states as the fundamental reality of the universe. Reality consists of matter, fields, spatial relations, and the motion and changes that actually occur. Physical states evolve according to natural laws, while time is a reference framework established by humans to record, compare, and calculate these changes. The paper adopts a fixed fundamental scale as one of its core hypotheses, represented by the constant parameter ΔC_0. System composition, interactions, and local conditions determine how this scale manifests in particular measurements. Subsequent models take the same ΔC_0 as an input to calculate transition intervals and relationships between readings under different conditions. On this basis, the paper distinguishes past, present, and future. The past comprises processes that objectively occurred; memories and records are possible traces of those processes remaining in present states. The present consists of physically existing states and ongoing processes. The future comprises subsequent changes allowed by the current state and natural laws but not yet realized. What is called the stopping of time is understood as the complete cessation of motion and state changes within a specified domain. The paper distinguishes Newton's conception of absolute time, the spacetime structures of special and general relativity, and the block-universe interpretation derived from spacetime geometry. It then uses thought experiments involving local stasis and repeated rewriting of history to examine claims about independent temporal passage and the accessibility of history, and discusses the problem of time that arises when combining quantum theory with general relativity. A "return" is defined as restoration to the state of a selected earlier moment. Data restoration, rejuvenation, reconstruction of scenes, and duplicates illustrate criteria for determining whether such a return has occurred. Finally, the paper examines the support that atomic-clock metrology, relativity, statistical physics, internal-clock research, and experiments on memory provide for its respective propositions. An additional empirical rule for comparing two clocks is introduced to demonstrate how a fixed parameter can enter a calculable result; its supplementary assumptions, calibration procedure, and parameter identifiability are made explicit.

Keywords: Space-State Evolution Theory; ontology of time; state-transition interval; fixed fundamental scale; absolute stasis; proper time.

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1. Introduction of Space-State Evolution Theory

The theory proposed here is formally named **Space-State Evolution Theory**, hereafter "the present theory." It takes space and the physical states within it as fundamental reality, treats motion as an important characteristic of states, holds that states evolve according to natural laws, and adopts the fixed fundamental scale ΔC_0 as one of its core parameters. Time is a metrological concept established by humans to compare physical changes; independently existing time has no ontological status.

1.1. There Is No Independently Existing Time, Only Space and Its Physical Changes

Coastal erosion, rock weathering, human aging, and the running of clocks each involve physical processes that can be investigated. Consider aging: the human body undergoes a sequence of physiological changes. Saying that a person has aged by ten years describes the interval associated with those changes using a human-chosen reference. The reading "ten years" supplies a scale of comparison. Why

these bodily changes occur should be explained through biological and physical mechanisms.

By “only space,” this paper means that a fundamental description of the universe begins with space and the physical states within it. Space here contains matter and fields, while a physical state includes the information needed to describe relevant processes, such as distributions of matter, motion, internal structure, and interactions. Objects occupying the same position may nevertheless differ in their internal states. This position assigns the explanation of all change to physical laws and denies that time exists as an independent physical entity. Time coordinates, seconds, and years, by contrast, are human-created means of measuring and describing objective changes.

Metrological references are selected by humans. Provided that a physical process is sufficiently stable, the number of its repetitions can be used to compare other processes. Counting a clock’s oscillations is one way to establish a correspondence between two physical processes. Humans define units, select clocks, and read their indications. Physical processes and the relations among them can nevertheless exist without observers.

Three levels of claims must be distinguished. First, units of time and timekeeping methods are descriptions of physical change. Second, this paper advances the stronger ontological proposition that time does not belong to the universe’s fundamental structure. For this proposition to become a physical theory, order, intervals, and clock differences should be reconstructed from spatial states and the laws governing their succession. Third, the paper separately assumes a single actual history and maintains that past and future have different modes of existence. The mere fact that time can be measured through physical processes does not by itself establish either of the latter two propositions. Section 1.5 specifies the fixed fundamental-scale parameter ΔC_0. Together with spatial states and the laws of their succession, it forms the theoretical starting point of this paper and is assumed in the subsequent derivations.

1.2. Definitions of Past, Present, and Future

The universe contains only physical reality as it is being realized. Processes that occurred earlier constitute the past; the existing state has been formed by processes that actually occurred; and the range of subsequent changes is constrained by natural laws and physical conditions. The three concepts are defined below, along with how each definition applies to the assessment of a particular event.

1.2.1. The Past Is the Physical Course of Events That Has Already Occurred

The past refers to motions, interactions, and state transitions that have already taken place. Earlier states once actually existed. Their phase as the current state of a system has ended. The paper therefore maintains that the past does not remain a physical location coexisting with the current state that can be entered again. The reality of history lies in the fact that the processes occurred.

Historical processes, historical states, and present traces must be distinguished. A historical process is the change that actually took place. A historical state is the physical state once possessed by a system at a particular stage of that process. A trace is a physical outcome preserved in present reality, such as a person’s memory, a photograph, an eroded rock, or an instrument record. What still physically exists

now is the traces and the media that carry them. They may provide information about earlier states, but may also be damaged, disappear, or support only incomplete inferences.

Suppose a landslide occurred on an unobserved hillside, the terrain was subsequently altered several times, and all associated traces disappeared. There would then be no observational basis for identifying the landslide. It would nonetheless remain part of actual history. Even if everyone lost the memory of an event, what changed would be present brain states and the historical information available. The paper thus treats whether an event occurred, whether it can presently be known, and how its records are represented in a model as three separate questions. Physical reality and the occurrence of historical events are both independent of human consciousness.

1.2.2. The Present Is the Physically Existing State of the Universe

The present refers to the physical states that actually exist and the physical processes that are occurring. Describing a person's present state includes their body, position, activities, and memories already stored in the brain. The content of a memory may refer to the past, while the neural state carrying that memory exists in the present. Describing the present state of a glass of water may involve its temperature, flow, surroundings, and ongoing exchange of heat. The present includes real motion and interactions. A pointer or coordinate on a time axis is only a device for labeling relevant states.

A present state may be the actual result of earlier changes and also the physical conditions confronting subsequent changes. It allows some components to continue changing while others remain unchanged. For example, while a person moves around a room, a shelf fixed to the wall may remain in the same position. Whether change has occurred depends on which physical quantities are being compared. Complete local stasis requires a stronger condition, as discussed in Section 1.3.

The phrase "the physically existing state of the universe" is a claim about what reality contains. For measurement purposes, states must be described through specific systems, events, and reference processes. To discuss whether distant events are simultaneous requires specifying a reference frame or a synchronization convention. Special relativity allows different inertial frames to assign different simultaneity relations to certain distant events [5, 7]. Accordingly, this paper initially defines the present through local processes and physically establishable connections. Any further claim that there is a global slice of the universe shared by all observers requires its own definition and physical justification.

1.2.3. The Future Consists of Subsequent States Allowed by the Present State but Not Yet Realized

Given a seed and its environment, natural laws constrain its interactions with water, air, and soil. It may germinate, or it may fail to germinate because the conditions are unsuitable. Which outcomes can follow depends on the system's state, external conditions, and relevant dynamics. The future consists of these as-yet-unrealized possibilities. In the present theory, there is no preexisting collection of future states waiting for reality to arrive at them page by page.

Two kinds of possibility must be distinguished. The first arises from incomplete information. Even if the actual dynamics would allow only one outcome given complete conditions, a person may make several predictions because not all conditions are known. The second arises from a particular probabilis-

tic model, which can assign probabilities to multiple subsequent outcomes under specified conditions. Whether an outcome can be predicted and whether it has actually occurred are separate questions. Even a uniquely predictable outcome still has to be realized through an actual physical process.

The paper additionally adopts the single-actual-history assumption. For any process that occurs, the outcome actually realized becomes part of reality and enters history. Other candidate outcomes can be used in probabilistic calculations or counterfactual analysis. This assumption accommodates uncertain predictions as well as deterministic laws. How a single outcome becomes realized in quantum theory, and how the corresponding probabilities are reproduced, still depends on a particular measurement model. A single actual history has ontological status within this paper and must be compared separately with relevant physical models.

1.2.4. How the Three Concepts Describe a Single Change

Table 1: Definitions of past, present, and future

Concept	Physical definition	Example: a glass of cooling water
Past	The course of events that actually occurred	The water was heated and has already transferred some heat to its surroundings
Present	The state that actually exists and the processes now occurring	The water has its current temperature and is exchanging energy with its environment
Future	Subsequent changes constrained by states and laws but not yet realized	The water may continue to cool or warm if heat is supplied externally

Suppose the water is heated back to its previous temperature. The same temperature appears again, but the preceding cooling process still occurred. Extending the comparison to more physical quantities makes the conditions for reproduction stricter; the act of restoration actually carried out in history remains a new operation. Chapter 4 applies this distinction to rejuvenation, reconstruction of historical scenes, and return to a state.

1.3. Stasis as the Absolute Cessation of Motion

This paper defines **absolute stasis** as a condition in which, throughout a course of events marked by an external physical process, all physical states under consideration within a specified region remain unchanged. “Absolute” here means that all changes within the scope of the freeze cease completely, including internal changes. To determine whether the region’s overall position and shape also remain unchanged, its spatial relations to surrounding objects must be specified. This definition does not require a reference frame of absolute rest for the universe.

Let A and B be two neighboring regions. The reference process in region B advances from reading C_1 to reading C_2, during which the state of B changes, while the state of A remains equal to some state S_A^0. This can be expressed as

$$S_A(C) = S_A^0 \qquad \text{for all } C_1 \le C \le C_2.$$

Here C is the reading of the reference process in region B. The definition requires the state to remain unchanged at every intermediate stage. If A changes and subsequently returns to S_A^0, the course of events still involves change, even though the initial and final states coincide. If only an object's position remains unchanged while its metabolism, oscillations, or interactions continue, only its position can be said to be unchanged. If all the constituents of a human body and their relevant physical states cease changing, metabolism, accumulation of damage, memory activity, and other physiological changes also cease; under that condition, aging does not continue. When absolute stasis is posited in a thought experiment, it remains necessary to explain separately which physical systems could attain it, and under what conditions.

1.4. Natural Laws Determine Which States Can Follow Other States

If states are observed in the sequence $A \to B \to C$, the question is how natural laws allow these states to follow one another. Saying that "time moves forward" can label the observed sequence. Explaining why this particular sequence occurs also requires interactions, initial conditions, and boundary conditions. A process may likewise have several possible outcomes. This paper adopts a **single actual history** as its basic position. Only one outcome actually occurs, although it may arise from probabilistic laws. The uniqueness of actual history and whether laws determine an outcome in advance are two different questions.

A local model can make this idea more explicit. Let S denote the physical state of a system, and let $R(S)$ denote the subsequent states permitted by natural laws and specified boundary conditions. $R(S)$ supplies a relation among candidate states, while interactions that actually occur constitute one particular course of events. For an open system, the conditions must include relevant exchanges of matter, energy, or information. This notation simply distinguishes the actual state from its permitted successors. After introducing the fixed fundamental scale of Section 1.5, the dependence can be written $R(S, L; \Delta C_0)$, where L collects specified interactions and local conditions. For the model to be predictive, the constraints, dynamics, and probability rules governing R must still be supplied.

1.5. The Fixed Fundamental-Scale Parameter and State-Transition Intervals

The paper adopts the hypothesis that **physical state transitions possess a fixed fundamental scale** as one of its core assumptions and consistently denotes the parameter by ΔC_0, stipulating that

$$\Delta C_0 = \text{constant}, \qquad \Delta C_0 > 0.$$

Within Space-State Evolution Theory, different systems and local conditions share the same fundamental-scale parameter. Its numerical value has to be established through a chosen measurement convention and experimental calibration. Throughout a derivation, ΔC_0 remains fixed; system composition, interactions, and local environments enter through other model inputs. Any explanations

and predictions concerning observations must follow from this premise.

To discuss a particular transition, suppose a system changes from state S_a to state S_b, and record the readings C_a and C_b of a reference process at the beginning and completion of the transition. This section denotes the observed interval by $\Delta C_{\text{obs}} = C_b - C_a$. ΔC_0 is the fixed fundamental parameter, whereas ΔC_{obs} is the difference between readings obtained under specified reference and experimental conditions. The former carries no location or system subscript; the latter can vary with the process and reference being compared.

Collect local conditions, system composition, and the driving protocol into L, and let P represent the reference and comparison procedure. The required metrological relation can then be denoted by

$$\Delta C_{\text{obs}}^{(P)} = \mathcal{F}_P(S_a, S_b, L; \Delta C_0).$$

Here $\mathcal{F}_P$ represents the correspondence jointly determined by specific dynamics and the measurement procedure. The equation makes explicit the distinct places occupied by the fundamental parameter and the experimental conditions. To obtain numerical predictions, the form of $\mathcal{F}_P$ must be derived from independently specified laws of interaction and then compared with experiment. Merely naming the function specifies only the inputs that the model ought to handle.

ΔC_0 denotes a fundamental scale for state transitions. By itself it neither requires transitions to occur in discrete jumps nor specifies the clock reading associated with any one complete transition. This paper takes the fixed parameter as a common starting point for subsequent models, leaving the dynamics to account for particular processes and how they appear in measurements. Physical references are still obtained by comparing processes; there is no need to attach to ΔC_0 another clock that flows autonomously throughout the universe.

Existing evidence indicates that some processes have stable properties as references while observed values can also depend on conditions [15–17]. Accordingly, this paper places experimental constraints on the relation between parameters and observations. A specific model must account for results under different conditions using the same fixed parameter and define its scope of applicability. Section 4.6 contrasts stasis, slowing, and restoration; Section 5.8 demonstrates a two-clock model taking ΔC_0 as an input; and Section 5.9 discusses tests across different timekeeping mechanisms.

1.5.1. The Relationship Between the Fundamental Scale and Time Measurement

The fixed fundamental-scale parameter ΔC_0 is used to describe an intrinsic scale associated with transitions between states of motion. Space-State Evolution Theory places it within the laws governing the succession of physical states in order to investigate how a specified transition occurs and how local conditions affect its measurable manifestation.

The observed interval of a transition, by contrast, is obtained from another physical process: for example, by counting how many cycles a reference clock accumulates while a particle completes a specified directional transition. The resulting difference in readings is denoted by ΔC_{obs}. The present theory requires this reading to be derived from the laws governing state transitions, the fixed parameter

ΔC_0, and specific physical conditions.

Thus, the theory first specifies physical states and their transition relations, and then establishes measurements through comparisons among processes. For this arrangement to constitute an independent explanation, it must specify the role of ΔC_0 in the laws of state transitions and show how ordinary measures of time arise from these relations.

2. Existing Theories of Spacetime and the Targets of This Paper's Critique

2.1. Newtonian Mechanics and Absolute Time

In the scholium following the opening definitions of the *Philosophiæ Naturalis Principia Mathematica*, Newton distinguished absolute time, which flows uniformly, from the ordinary measures of time obtained through motion. Days, months, and years are means of measurement. Absolute time, in contrast, is understood to possess its own uniformity, with duration independent of those measurement processes. In the usual formulation of classical mechanics, two systems can be described using a common time parameter. How they move is nevertheless determined by their respective dynamics and initial conditions [1].

Both positions acknowledge that motion can be compared using clocks. Newton's conception of absolute time understands duration as an independently existing background to which measurement refers. This paper instead attempts to explain duration directly through relations between processes. Section 3.1 asks whether independent temporal passage offers an additional explanation for why two systems undergo different changes.

2.2. Special Relativity and Proper Time

Special relativity describes motion in terms of events and their spacetime relations. Different inertial frames may assign different simultaneity relations to certain distant events. For events that can be connected by light signals or slower signals, causal order is constrained by spacetime structure. Relativity thus preserves an objective causal ordering while abandoning Newtonian universal absolute simultaneity [5, 7].

Coordinate time supplies labels for events within a reference frame. Proper time accumulates along the timelike path followed by an ideal clock. In an inertial frame in flat spacetime, if a clock moves at speed v relative to that frame, then $\mathrm{d}\tau = \mathrm{d}t\sqrt{1 - v^2/c^2}$. When two clocks separate and subsequently reunite, their accumulated readings can be compared directly. If this paper is to derive these measurable relations from spatial states, it must explain why relative motion produces the corresponding differences in readings and calculate their magnitudes [5, 7].

2.3. General Relativity and Dynamical Spacetime

General relativity uses the spacetime metric to describe local intervals, light propagation, and proper time. The distribution of matter and energy is related to spacetime geometry through the field equations. Under the $(-,+,+,+)$ sign convention and with $x^0 = ct$, proper time along a timelike path is given by $\mathrm{d}\tau = c^{-1}\sqrt{-g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu}$ [7, 13]. This makes specific clock differences predictable. Section 5.8

gives a quantitative benchmark for stationary clocks at different heights near Earth's surface.

Some mathematical solutions of general relativity permit closed timelike curves. In such models, a timelike path can return to the same spacetime event. Whether these paths can form in reality, which events they can reach from a given location, and how history would satisfy consistency conditions all depend on the specific model. The cited billiard-ball study analyzes self-consistent trajectories in a model that presupposes a wormhole. Hawking investigated possible mechanisms that would prevent the formation of closed timelike curves and proposed the chronology protection conjecture [2, 3]. The ability to return arbitrarily to a specified event and rewrite what follows is therefore a stronger proposal that requires separate examination.

2.4. The Block-Universe Interpretation and the Accessibility of History

A four-dimensional spacetime model can represent the geometric relations among different events within a unified framework. Putnam, for example, advanced an argument from the relativity of simultaneity that past, present, and future events are equally real [14]. This concerns how events exist within the model and is commonly classified as a block-universe interpretation. The paper's definitions of past and future differ substantively from this interpretation. Here, past events genuinely occurred, current reality consists of actually realized states, and future changes have yet to be realized through physical processes.

Even if a block-universe interpretation is adopted, whether a particular event can be reached from a given location still depends on whether a physically lawful path exists. Even if some return path is permitted, changing the original event and then entering the altered subsequent history requires additional rules.

Table 2: Comparison with existing theories of spacetime

Object of comparison	What it asserts or describes	The disagreement addressed here
Newtonian absolute time	Duration is independent of measurement processes; classical dynamics may employ a shared time parameter	Whether explaining duration and change requires an independent temporal background
Special and general relativity	Spacetime structure describes causation, paths, and proper time	Whether these measurable structures can be derived from relations among spatial states
Block-universe interpretation	Past, present, and future events have equal reality	Whether what has occurred, what is occurring, and what has not yet occurred have different ontological status
Models of revisitable and rewritable history	Assume that events can be revisited, altered, and followed by a rewritten subsequent history	How such paths are possible, how alterations connect to subsequent states, and how outcomes are selected

3. Questions About Independent Time and Revisitable Spacetime

The following thought experiments ask what common temporal passage adds to an explanation of local change, whether history that has already occurred can be treated as a place presently reachable, and whether arbitrary return, alteration, and revisitation can be explained consistently within a single model. The assumptions targeted by each critique differ. Only by separating them can one determine under which theoretical interpretations a particular objection holds.

3.1. How Local Stasis Fits into a Common History

In the scholium of the *Principia*, Newton distinguished uniformly flowing absolute time from ordinary time measurement obtained through motion [1]. The first challenge posed here to this ontology is as follows: If neighboring regions A and B are under the same external conditions, while the physical state of A remains unchanged and B continues to change, what does their purportedly shared passage of time explain about the difference?

Both regions can be compared using the same clock in region B. For example, the reference clock in B advances by ten seconds, the specified state of A remains unchanged, and B continues to evolve. History can accurately contain both facts. Sharing a measuring scale requires only a common reference; it does not require two systems to change, let alone to change at the same rate. Stopping a clock in A directly changes only how that particular clock functions. If a clock moved into A continues to run, its internal state is still changing. The system originally under discussion in A may nevertheless remain stationary; however, if the newly introduced clock is included in the frozen domain, it must be reassessed whether the entire region satisfies the definition of absolute stasis.

This example calls for a distinction between metrological parameters and dynamical causes. A theory that uses a common time parameter can have both constant and changing solutions, determined respectively by the relevant physical conditions and laws. The question raised here is: If those conditions and laws already perform the calculations needed to explain change, what additional calculable content does the ontological claim of independent passage supply? Space-State Evolution Theory attempts to establish measurements of duration and rates through state relations as well. Completing that task requires explicit derivations of the ratios and conversion rules between processes.

3.2. How Can a Past That Originally Had No Time Traveler Accommodate One?

Suppose a person, at age fifty, attempts to return to a particular location in the year when they were twenty, retaining the body and memories of their fifty-year-old self. If the original history contained only the twenty-year-old person at that location, the arrival of the fifty-year-old changes the physical state of the scene. If that person also intervenes in the actions of others, the affected subsequent events must be reinterpreted accordingly.

The objection can be framed as a conditional contradiction. Let S be the original complete state of the scene, and S' the state after the traveler arrives. If "returning to the original past" requires the complete reproduction of S while simultaneously requiring a traveler who was not originally there, the conditions $S' = S$ and $S' \neq S$ are imposed at once. They cannot both hold. Whether the traveler was

already part of the original past is crucial to this objection.

In a fixed-history model, the fifty-year-old traveler may have been part of that history all along, and the simultaneous presence of the twenty- and fifty-year-old selves at the same scene would then agree with the model's premises. Certain solutions of general relativity permit closed timelike curves, and studies of self-consistent billiard-ball trajectories demonstrate possibilities within such models [2, 3]. These models must include the traveler's appearance in the original history. A claim that the traveler was absent from the original history but can subsequently be added requires a further explanation of how history is rewritten and how the original and rewritten histories are related.

3.3. The Cosmic PowerPoint Analogy and Repeated Rewriting of History

This thought experiment begins by separating four premises. First, time exists independently. Second, past events continue to exist in some sense. Third, there is a physical path from a present event to a specified past event. Fourth, after arriving, one can alter an established event and enter and revisit the subsequent history generated by the alteration. Each earlier premise does not by itself entail the next. The PowerPoint analogy chiefly tests the combination of the third and fourth premises. Suppose history is a PowerPoint presentation in which every slide represents the state of a particular stage, while a traveler can supposedly flip to any slide and alter its contents. Someone returns to a junction yesterday. Originally they went straight ahead; upon returning they turn instead. A model permitting this alteration must supply a subsequent history consistent with the turn.

If the traveler can revisit the same slide to try a third action, and can return to a slide in a previously modified history to change it again, each set of subsequent events may become the starting point of another alteration. Even if there are only finitely many choices at each step, repeated operations can generate different subsequent histories without a fixed upper bound on the number of candidate histories.

This objection targets models that jointly permit **arbitrary return, alteration of established events, and revisitation of rewritten outcomes**. If every version is preserved and accessible, the model must explain how the target version is identified and how an operation of return or rewriting connects to a particular history. If older versions are overwritten, it must specify the overwriting rule and how the traveler's retained body and memories remain compatible with the altered course of events. Even if a model can mathematically represent infinitely many candidate histories, it must still account for which ones are actually realized and which outcome corresponds to a particular operation.

If the model additionally stipulates that each permitted outcome becomes an actually existing, permanently preserved branch, then each branch containing the traveler also contains its own actual version of that traveler. Given infinitely many such branches, the model is committed to the existence of infinitely many traveler versions. Repeated rewriting increases the number of histories and interpersonal relations that need to be distinguished. Such counting must distinguish versions in different branches from the different age stages that appear in one scene; the two are determined by their respective physical histories.

A more specific objection follows. If a traveler intends to return purposefully to an event, the target

must identify both the version of history and the event within that version. Location and date alone may still correspond to several distinct scenes. The model must specify how the current state of the device, the input information, and boundary conditions determine the arrival event. If multiple outcomes are allowed, it must specify a selection rule or probabilities. If further returns are permitted, the same rules must explain how later arrivals connect with earlier traversals and versions. Saying that the original history already contained a traveler answers a question about headcount but does not yet specify these mechanisms of localization and succession.

Fixed-history, overwriting-history, and preserved-branch approaches therefore carry different requirements. Fixed history requires every arrival to belong to the same self-consistent course of events and cannot at the same time promise arbitrary alteration of established events. Under that assumption, recounting the departure that led to an arrival already included in history does not create another person; additional age stages require additional actual arrivals. An overwriting model must define how old versions are replaced and the provenance of information retained by the traveler. A preserved-branch model must define how versions persist, can be accessed, and relate to one another. Any model promising arbitrary return, arbitrary rewriting, and the preservation and accessibility of all outcomes must handle these requirements consistently. Infinitely many versions do not by themselves constitute a logical contradiction; the central question is whether all these capabilities can be implemented by a single set of calculable physical rules.

The billiard-ball study offers a narrower point of comparison. In a particular wormhole model, certain specified initial trajectories can admit infinitely many self-consistent solutions [2]. This shows that, under some conditions in that model, the given initial information does not uniquely determine what follows. It is therefore necessary to distinguish **mathematically permitted solutions, outcomes that actually occur, and accessible histories**. The PowerPoint analogy goes further by asking whether a model that treats arbitrary rewriting as a genuine capability can provide an explicit physical definition for each operation.

This paper assumes a single actual history. Current conditions can permit multiple candidate outcomes, while the processes that actually occur determine how that particular history develops. Records of changes that have already occurred can be preserved and used to guide later restoration of a state. Chapter 4 discusses the criteria for such a return.

3.4. External Time Parameters and Internal Physical Measurement

Conventional nonrelativistic quantum mechanics uses an external parameter t to describe the evolution of a quantum state. For example, the Schrödinger equation is $i\hbar\,\partial_t|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$. General relativity, by contrast, relates clock readings to paths in dynamical spacetime. When these theories are combined, choosing a reference for evolution, constructing observables, and maintaining the corresponding constraints gives rise to the problem of time in quantum gravity [18].

This problem is adjacent to the research direction pursued here. Within a closed system that also includes the clock as an object of study, an internal variable C may be chosen as a reference and the conditional relations between it and other physical quantities calculated. For example, one can

investigate the conditional probabilities of different values of a quantity Q given a particular reading of C. The internal-clock research listed in Section 5.4 demonstrates specific constructions along these lines [10, 18].

Space-State Evolution Theory begins with state succession and the relations used to measure it, and requires a selected internal reference and the fixed fundamental parameter ΔC_0 to enter a specific model together. The remaining difficulties become explicit tasks: specifying when the reference is usable, handling interactions that affect the reference itself, checking whether observables remain consistent when another reference is chosen, and recovering conventional quantum probabilities and relativistic approximations. This paper has not yet supplied a unified solution to these tasks. The role of the fixed fundamental-scale assumption must be tested in such constructions.

4. Interpretation and Examples of Space-State Evolution Theory

This paper defines a "return" as **restoration to the state of a selected earlier moment**. Determining whether the return succeeds requires specifying the object, the target state, and the scope of restoration. The selected moment identifies the state to be reproduced, while restoration is accomplished through a present physical operation. The following discussion begins with data recovery, for which the criteria are clear, and then examines the human body, reconstructed scenes, and limiting cases of restoration. It concludes by using two-clock comparisons and the contrast among stasis, slowing, and restoration to clarify rates of change and transition intervals.

4.1. Understanding Return to a State Through Lossless Decompression and Data Migration

Consider electronic data. The contents of a file are first encoded and stored in a compressed archive. If a lossless encoding is used, correct decompression can restore the original file byte for byte [12]. Suppose an old version of a document is saved and the document is subsequently edited. Decompressing the saved version and writing it back returns the document's contents to the saved state. This provides a concrete criterion for a "return." If the restored contents match the specified version, the document can be said to have returned to that state. The editing and restoration remain separate operations that actually occur in succession.

Data migration illustrates another aspect of the same principle. Logical data can be represented by zeros and ones, but different storage media may encode them physically in different ways. Moving a file from one medium to another preserves and reconstructs the same content if it yields identical data when read according to each medium's rules. Migration verifies the identity of the content. If the migration target is an earlier saved version, it can also restore the data state corresponding to that saving operation. This makes it possible to specify more precisely "what has been restored." If only the document's contents must match, the medium and its encoding may change. If the file's additional attributes must also be restored, they must be included in the recorded information and the verification criteria.

To apply this idea to the thought experiment, let S_0 be a person's target bodily state in youth, or the target physical state of a region at a selected moment. Let I_0 be a record describing the specified

scope of restoration. With E denoting encoding and D decoding, lossless recording at the information level requires $D(E(I_0)) = I_0$. Decoding restores only a description of the state. Returning a person or a region to the state also requires physical reconstruction based on that description, followed by verification that the reconstructed result meets the state criteria of S_0.

For a person, restoration might cover only appearance, or it might hypothetically cover internal bodily structure and the states carrying memories. The former is a partial return; the latter corresponds to complete bodily-state restoration in the thought experiment. For a region, restoring its former physical state requires that the record cover the matter distribution, internal motion, and fields of interest, and that the treatment of links between the region and its surroundings be specified. If the record contains only visible appearance, decompression and reconstruction can reproduce only the corresponding aspects of the state. If replacing some materials or carriers is permitted, the structural, behavioral, or measurable properties required to remain identical must be stated. If the target requires microscopic states to match to a specified precision, the criterion must cover the corresponding degrees of freedom. The analogy with decompression explains what restoration to a state means. The ability to read, store, and reconstruct the entire physical state of a person or a region remains an assumption of the thought experiment.

If the physical state within the specified scope is indeed restored, the object may be said to have returned to the state of the selected moment. Equality between two states is a judgment about the outcome of reconstruction. The original appearance of the state and its later restoration remain distinct events within a single historical course.

4.2. Rejuvenation Is an Operation on the Present Body

Suppose that, at age eighty, a person could reconstruct their bodily state from information saved when they were twenty. Restoring only the skin, appearance, and some physiological indicators could produce a person with youthful bodily features while retaining later experiences and memories. If the thought experiment further assumes that all physical states relevant to memory and cognition are restored precisely, then, on the assumption that mental life depends on the relevant physical states, later-acquired memories would also be altered together with the physical states that carry them. The extent of the change depends on which structures, activities, and relations the restoration covers.

Both operations take place in existing physical reality. The equipment used, the operation itself, and external records can preserve the causal provenance of the restoration. Even if the body's state matches perfectly within the specified scope, the experience at age twenty and the restoration at age eighty remain two distinct historical events. Section 5.5 discusses experiments supporting the dependence of memory on physical states. The feasibility of complete human restoration, however, is an assumption of the thought experiment.

4.3. Reconstruction of Historical Scenes and Duplicates

Imagine reconstructing a street as it was in the past from recorded data, including buildings, sounds, people, and every state within that space. If someone who previously visited that street enters the

reconstructed scene and temporarily lacks external information, they may mistake the scene before them for the one they once experienced. With sufficiently precise reconstruction, the person's experience may come close to their memory of the original scene. Determining which event the scene belongs to also requires examining its materials, its construction, and its causal connections with the surrounding world.

The example of a duplicate tests this point further. Suppose two individuals have the same appearance, brain states, and memories within the scope under consideration. They can occupy different positions, receive different information, and develop their own experiences. Similarity judged from a present state and historical provenance judged from the course of events are two different kinds of information. Constructing a sufficiently precise duplicate within the existing universe can yield a real individual with the corresponding properties. That individual's identity and experiences must also be discussed in terms of the process by which they came to exist.

4.4. Complete Restoration and the Distinguishability of Historical Events

A stronger thought experiment requires restoring the object, its environment, instruments, and all available records simultaneously. If every measurable difference has been excluded by assumption, an observer within the system cannot distinguish the original occurrence from its reproduction by those state readings. This paper nevertheless understands them as different occurrences within the actual course of events. Consider $A \to B \to C \to A'$. If A' and A have identical measurable properties, those properties alone cannot show whether the system entered A through its initial preparation or reached A' after passing through B and C. This also poses a research problem for the present theory: How can a model retaining only the current state represent a state's historical provenance? If an additional historical relation is needed, does it have observable consequences?

Placing this case alongside the PowerPoint thought experiment highlights the different mechanisms the two proposals require. State restoration must explain how to modify a current system so that it satisfies a historical description. A model permitting return to and alteration of an established event must additionally explain how the original event is reached, how the altered sequence continues, and how different histories correspond. The two arrangements may produce similar subjective experiences but impose different explanatory demands on physics.

4.5. Gravitational Environments and the Relative Rates of Physical Processes

Suppose two identical light clocks are placed at fixed locations near the ground, one higher and one lower, and their frequencies are compared by light signals. The clocks calibrate their oscillating signals against atomic resonances and accumulate cycles to generate readings [15]. Using the lower clock A as a reference, when it accumulates ΔC_A timing cycles, a specified comparison procedure determines the number ΔC_B accumulated by the higher clock B over the corresponding course of events. Their relative rates of change can be written as $\Delta C_B / \Delta C_A$. In this experimental arrangement, the higher clock runs faster relative to the lower clock. Both clocks may remain stationary relative to the ground [6].

In the present theory, the expression “different rates of the flow of time” means **different rates at which two physical processes complete state changes under the same comparison procedure**. “Faster” and “slower” here refer to relative rates of internal timekeeping processes, such as how many cycles one clock completes while a reference clock completes a specified number. The speed at which a clock’s body moves relative to the ground is a different physical quantity. When each clock is compared locally with an identical ideal clock at the same place, normal local timekeeping proceeds as usual. The difference emerges when the processes at the two locations are compared through signals [6, 7].

In a weak, stationary gravitational field, the frequency comparison between stationary clocks depends on the difference in gravitational potential between their locations. Near Earth’s surface, gravitational potential is higher at greater height, and the higher clock runs faster relative to the lower one. The magnitude of gravitational acceleration and gravitational potential are different quantities. Even when gravitational acceleration is approximately equal at two locations, their different positions can still correspond to a potential difference and a measurable clock difference [6, 7]. The task for this theory is therefore to explain how a given gravitational environment constrains the rate relation between two physical processes.

Existing relativity also provides an analogy involving motion. Consider a laboratory cabin accelerating upward. By the time light emitted from its floor reaches the ceiling, the receiver has acquired additional velocity relative to the instantaneous inertial frame at the moment of emission. An analysis in that inertial frame finds that the signal received at the ceiling is Doppler redshifted. By the equivalence principle, this analogy illustrates the direction of frequency shifts in a weak gravitational field [7]. It also shows why a comparison protocol must include emission, propagation, and reception of signals, together with their physical conditions.

Under Section 1.5, the processes at both locations use the same fixed fundamental scale ΔC_0. Differences between the numbers of cycles result from the way gravitational conditions, the physical composition of the clocks, and the comparison procedure enter the metrological relation. To make this account predictive, it is necessary to derive the relation between these inputs and the ratio of cycle counts, and to explain why different kinds of ideal clocks yield corresponding comparisons. Section 5.8 provides a quantitative benchmark that predictions using ΔC_0 must reproduce.

4.6. Unified Criteria for Local Stasis, Slower Change, and State Restoration

Choose an external reference process with readings increasing from C_a to C_b. Let $S(C)$ represent the physical state of the human body as compared with that reference, and let S_{target} be the specified restoration target. Three kinds of operation can be distinguished over the same reference interval.

Table 3: Criteria for local stasis, slowing, and state restoration

Operation	State criterion	Bodily change and historical records
Complete cessation of all relevant changes	$S(C) = S(C_a)$ for every reference reading in the interval	Metabolism, accumulation of damage, and other physiological changes within the assumed scope all stop, while the external reference continues to change
Slowing of a specified process	A specified measure of progress accumulates less than in a control process under comparable conditions, although the body continues to change	The affected physiological processes must be identified individually; external records can continue to accumulate
Restoration to a specified state	After the operation, $S(C_b) = S_{\text{target}}$ according to the specified restoration criterion	The body reaches the target state; both the preceding experiences and the restoration operation have occurred

"Slowing" requires a measurable quantity for comparison, such as the accumulated count of a reaction or an indicator of damage. In a complex biological system, a smaller change in one indicator does not by itself imply that every aging process has slowed by the same proportion. If the thought experiment stipulates that all relevant changes cease, then no physiological aging occurs during the freeze. This conditional conclusion must be considered separately from any physical mechanism that could implement such freezing.

These criteria represent three distinct operations within Space-State Evolution Theory. Stasis requires constancy throughout the entire reference interval; slowing requires a reduction in the relative accumulation of a specified measure of progress; and restoration requires a match to the target state at the end of the operation. All three can be described through comparisons between states and reference processes. Specific dynamics, using the fixed parameter ΔC_0, the system state, and local conditions as inputs, must then establish which operations can occur and what records they leave behind.

5. Support from Existing Theories and Experiments

5.1. Resonance Frequencies, Transitions, and Accumulated Readings in Atomic Clocks

The International System of Units defines the second by fixing the numerical value of the unperturbed ground-state hyperfine transition frequency of the cesium-133 atom at 9,192,631,770 Hz [4]. Atomic clocks use atomic resonance responses to calibrate an oscillating signal and then count its cycles. Practical implementations also involve physical operations such as state preparation, detection, and correction [15].

When discussing the fundamental scale, three quantities must be distinguished.

Table 4: Distinct quantities relevant to atomic clocks and state transitions

Quantity	What it describes	Role in investigating a fundamental scale
Atomic resonance frequency	The oscillation frequency corresponding to a specified energy-level difference	Provides a frequency reference for comparable processes under specified conditions
Rate of driven transitions	How rapidly the state of a system changes under a given driving interaction	Tests the relation between driving conditions and transition intervals
Accumulated clock reading	How many cycles are completed by an oscillating signal calibrated against atomic resonance	Uses one physical process as the metrological reference for another

Atomic energy levels can be influenced by electromagnetic fields, thermal radiation, and other conditions; these effects must be measured or corrected in the evaluation of clock frequencies [16]. The rate of a driven transition also depends on the driving interaction. In a coherently driven two-level model, for example, the Rabi frequency characterizes the scale of the driven state change and depends on the coupling field [17]. Resonance frequency and driven-transition rate must therefore be identified separately in experiments. The cycles counted by an atomic clock and the interval required to complete a specified state transition have different physical definitions.

These results provide experimental and theoretical grounds for stable physical references and their dependence on conditions. This paper takes the fixed parameter ΔC_0 as a common input and requires a specific atomic model to establish separately the relations among energy-level structure, driven transitions, and accumulated readings. Selecting a known frequency as a reference establishes a measure. Calculating the dependence of these quantities on ΔC_0, however, requires explicit dynamics. Comparing different processes can test whether derivations using the same parameter agree with observation.

5.2. Relativity Relates Measurable Clock Rates to Physical Paths

Special relativity distinguishes the simultaneity of distant events in different reference frames from the proper time accumulated along a clock's own path [5]. Optical atomic-clock experiments have also measured the fact that a stationary clock at a higher location in a terrestrial laboratory runs faster, and that changes in motion can produce measurable frequency differences [6]. These results support the paper's questioning of a universal absolute rate of temporal passage and furnish objective facts that must be explained. Which clock humans choose as a reference is conventional, while the observed differences between two clocks are constrained by physical conditions.

Consider a nonrotating black hole. When two ideal clocks are held at fixed radii outside the horizon and compared according to a specified procedure, the clock nearer the black hole runs more slowly relative to the more distant one. The rapid motion of nearby gas describes the state of the gas and cannot substitute for a calculation concerning the two stationary clocks. A falling clock follows a different path, and standard relativity allows it to cross the horizon in finite proper time [7]. If Space-

State Evolution Theory is to offer an independent explanation, it must reconstruct the magnitudes and directions of these clock differences and their dependence on paths and gravitational conditions. An unchanged material state and zero proper time must also be modeled separately.

5.3. Statistical Physics Provides a Mechanism for the Direction of Macroscopic Change

To explain why the universe changes in the sequence $A \to B \to C$, one can begin with specific physical processes. For example, statistical physics relates the usual macroscopic direction of cooling water to the statistical properties of microscopic states and to low-entropy initial conditions [8]. If a room catches fire, the water may heat up again. The heat source changes the system's boundary conditions and must be included in the explanation. This example supports the present approach: when a sequence or direction changes, the explanation should examine how interactions and physical conditions change.

Entropy increase can explain the direction of many macroscopic processes. Explaining why a particular event follows another also requires the corresponding dynamics. Microscopic changes can continue even in a system at equilibrium, so constant macroscopic entropy cannot simply be equated with absolute stasis.

5.4. Internal-Clock Models Show How Time Descriptions Can Be Built from Relations Among Processes

In some formulations of classical dynamics, one can first describe a path through configuration space and then use dynamical relations to determine an effective time along it. Barbour discussed the possibility of this approach in classical dynamics [9]. In quantum theory, Page and Wootters examined how correlations between a clock and an observed system in the overall state can be used to represent conditional states of the system at different clock readings [10].

Their common ground with this paper can be expressed as a concrete task: given a clock process and a process under study, calculate which states the latter can occupy, and with what probabilities, when the clock has a particular reading. These works provide theoretical support for such approaches within specified models. Further work must identify which physical relations may serve as fundamental inputs and when the models recover conventional descriptions of time. If an internal clock's reading stops changing, one must determine whether another physical reference should be used or whether all relevant states of the system itself have genuinely remained unchanged.

5.5. The Physical Basis of Memory Supports the Distinction Between State Restoration and History

Experiments in mice have found that activating specific populations of hippocampal neurons tagged during learning can elicit behavior consistent with retrieval of the associated fear memory [11]. This provides experimental support for the dependence of memory retrieval on the state and activity of physical systems. The result plays a different argumentative role from the thought experiment about rejuvenation. The experiment demonstrates that manipulating neural activity can affect memory-related

behavior. The thought experiment goes further and asks whether, if memory-related states could be restored in their entirety, the memories subjectively available to a person would establish that the person had returned to the original event. The identity of a historical event still has to be assessed through the course of its occurrence and its external connections.

5.6. A Calculable Expression for Physical References

Let Q be a physical quantity of the system being measured and C the reading of a reference system. Given a fixed fundamental parameter ΔC_0 and specified local conditions L, the correspondence between the two processes can be denoted by $Q(C; L, \Delta C_0)$. On an interval where C is continuous and strictly monotonic, $Q(C)$ can describe their correspondence. If a differentiable auxiliary label λ is used to describe this course of events, then wherever $\mathrm{d}C/\mathrm{d}\lambda \neq 0$,

$$\frac{\mathrm{d}Q}{\mathrm{d}C} = \frac{\mathrm{d}Q/\mathrm{d}\lambda}{\mathrm{d}C/\mathrm{d}\lambda}.$$

Under a relabeling that preserves order and has an everywhere-positive derivative, the numerator and denominator acquire the same transformation factor, so the ratio remains unchanged. It expresses the change in the measured physical quantity per unit change in the reference reading. If Q remains constant throughout the interval, that physical quantity can be judged unchanged over the interval. Absolute stasis requires this assessment for all relevant physical states within the specified scope.

Such measurement depends on the reference process being able to distinguish the stages under investigation. A periodic clock with only a dial reading and no cycle count can display the same reading repeatedly. If the reference process itself stops, another physical process capable of distinguishing the stages must be used. Relabeling does not alter the measured correspondence. Determining which correspondences are physically realizable additionally requires dynamics. The example provides a metrological method using an internal reference and makes the derivation of clock differences from concrete laws a separate task [9, 10].

5.7. Connections to Relational Theories of Time and Research on Quantum Time

This paper shares with relational theories of time the idea that measurement can be understood through relations among physical processes. Beyond that, it advances its own ontological claims concerning past, present, and future, and combines a single actual history with criteria for absolute stasis. Relational measurement can occur in models with different views of history. Even where two models use internal clocks to describe change, they may answer differently whether the past still exists in some way and whether the future has already been realized.

Internal-clock and emergent-time research investigates whether the time structure normally written into equations can instead be obtained from more fundamental physical relations. Classical paths through configuration space and quantum conditional states offer specific examples [9, 10]. In a quantum model, however, explaining the formation of a single actual history also requires dynamics of the measurement process and probabilities for its outcomes. Information obtained by people and instruments is part of a physical process. Whether consciousness must play a role and whether only one

outcome is actually realized must be addressed separately within the chosen theory. The internal-clock construction itself supplies a way of defining references for such discussion.

5.8. A Two-Clock Comparison Model with a Fixed Fundamental Scale

5.8.1. The Weak-Gravitational-Field Benchmark to Be Reproduced

Consider two identical ideal clocks held stationary near Earth's surface at heights separated by Δh. Their frequencies are compared by means of a specified signal protocol. Let A be the lower clock, B the higher one, and Φ the gravitational potential in the weak-field approximation. Neglecting relative motion, instrumental perturbations, and higher-order corrections, standard relativity gives

$$\frac{\nu_B - \nu_A}{\nu_A} \simeq \frac{\Phi_B - \Phi_A}{c^2} \simeq \frac{g\,\Delta h}{c^2}.$$

With $g = 9.8\,\mathrm{m/s^2}$ and $c = 299\,792\,458\,\mathrm{m/s}$, a height difference of one meter yields an expected fractional frequency difference of approximately 1.09×10^{-16}, with the higher clock running faster. This is an established theoretical benchmark that the model in this section must confront; related frequency comparisons have already been performed with optical clocks [6, 7].

5.8.2. An Explicit Supplementary Comparison Rule

Section 1.5 used $\mathcal{F}_P$ to indicate how states and readings depend on one another. To provide an example that can actually be calculated, this section makes an additional empirical assumption. Suppose all readings have first been converted into a consistent nominal unit. Let

$$\frac{\mathrm{d}C_B}{\mathrm{d}C_A} = r_{BA} = \exp\left[\frac{\chi}{\Delta C_0}(\Phi_B - \Phi_A)\right].$$

Here ΔC_0 is the theory's fixed fundamental scale, and χ is an additional gravitational-potential coupling coefficient introduced for this example. The ratio $\chi/\Delta C_0$ has units of inverse gravitational potential, making the exponent dimensionless. In a particular version of the example model, both parameters remain fixed, while the difference in the physical conditions at the two locations determines the variation of the reading ratio. This exponential rule is a supplementary modeling assumption chosen for this section; it has not yet been derived from the laws governing state succession.

The chosen form permits direct checks of three consistency conditions. When the gravitational potentials at the two locations are equal, the reading ratio is one. Swapping the reference clocks gives the reciprocal ratio. Within the same static comparison arrangement, a third clock D can be used as an intermediary, giving $r_{BA} = r_{BD}r_{DA}$. These properties allow results to be converted consistently. They test only the mathematical structure of this static comparison rule; signal propagation, motion, and more general gravitational environments still require explicit modeling.

If the clocks are held fixed, the gravitational environment is stationary, and the comparison protocol has been specified, then r_{BA} is constant. Taking C_A as an internal reference and integrating gives

$$C_B(C_A) - C_B(C_{A0}) = r_{BA}(C_A - C_{A0}).$$

The equation directly calculates how much one clock should accumulate when the reference clock advances by a specified amount. It does not introduce an additional universal parameter of temporal passage.

5.8.3. Numerical Example Following Calibration

Write $\kappa = \chi/\Delta C_0$. In the regime $|\kappa(\Phi_B - \Phi_A)| \ll 1$, the model gives $r_{BA} - 1 \simeq \kappa(\Phi_B - \Phi_A)$. If $\kappa = 1/c^2$ is calibrated against the standard relativistic weak-field result, the following examples can be calculated.

Table 5: Calculated two-clock reading differences for different height separations

Height difference Δh	Calculated fractional reading difference $r_{BA} - 1$	Additional reading accumulated by the higher clock during one day on the lower reference clock
0.33 m	3.598×10^{-17}	Approximately 3.11 ps
1 m	1.090×10^{-16}	Approximately 9.42 ps
10 m	1.090×10^{-15}	Approximately 94.21 ps

One day in the table means 86,400 seconds on the reference clock; one picosecond (ps) is 10^{-12} seconds. The figures are calculated using the specified values of g and κ to show how the rule produces comparable results. They are not being reported as new experimental measurements. Calibration using an existing weak-field result can reproduce the scale of that result. For an independent prediction, the rule and its parameters must first be fixed and then tested against observations not used for calibration. Conclusions beyond this static weak-field setting require separate justification.

5.8.4. The Role and Identifiability of the Fixed Parameter

When comparing candidate models with different parameter values, if χ is fixed independently, different values of ΔC_0 yield different values of r_{BA}. In each specified model, ΔC_0 remains constant. The influence of the fundamental parameter on predictions can thus be tested. However, these two-clock ratios by themselves determine only $\kappa = \chi/\Delta C_0$. Multiplying χ and ΔC_0 simultaneously by the same positive number leaves all predictions unchanged.

This result identifies what is still missing before the example model can provide a measurement of the fundamental scale. Determining ΔC_0 and χ separately requires an independently specified relation for the coupling coefficient or a second observable with a different dependence on the two parameters. Merely introducing an additional named parameter does not ensure that it can be measured independently.

The general function $\mathcal{F}_P$ in Section 1.5 should be examined in the same way. If every new dataset can be matched by freely adjusting the function, the model does not yet constrain outcomes. Future work should specify its functional form, fixed parameters, and domain of application, and state which observations would require their revision. What has been established here is the role of the parameter within an additional model; the hypothesis of a fixed fundamental scale itself still needs support from dynamics and independent tests.

5.9. Consistency Tests Across Different Timekeeping Mechanisms

A single type of clock can be used to test how it changes under different conditions. If the present theory requires the same ΔC_0 to describe different timekeeping mechanisms, relations among those mechanisms should also be compared. First, convert the characteristic nominal frequencies of different clocks into consistent readings; then examine whether changes in local conditions produce the relative responses stipulated by the model.

Long-term comparisons of hydrogen masers and cesium clocks have tested whether the ratios of two atomic frequencies respond differently to variations in the Sun's gravitational potential. The results are consistent with zero differential response and place limits on the size of any such difference [19]. Models attributing gravitational effects to particular timekeeping mechanisms must therefore calculate mechanism-dependent differences and compare them with the corresponding experimental bounds.

In the example of Section 5.8, one can additionally stipulate the same χ for all kinds of ideal clocks. They would then exhibit the same normalized response to a given gravitational-potential difference. This universal-coupling stipulation has a clear experimental meaning, but its origin still needs to be explained through the dynamics of the different clocks. Alternatively, assigning different coefficients χ_i to different clock types would yield testable mechanism dependence. The rules for either version must be specified before comparison.

The roles of experimental factors must also be clear. Systematic biases of measurement apparatus require evaluation; an externally applied drive is an input to the model; and the gravitational environment and internal system interactions are conditions that the theory must describe. Initial states, comparison methods, parameters, and uncertainty ranges must be fixed in advance, with independent data used for testing. If reproducible results under controlled conditions fall outside the predictions and uncertainties of a model version, that version should be revised or abandoned. Any proposed new influencing factor must come with independent evidence and measurable consequences.

6. Conclusion

Space-State Evolution Theory maintains that independently existing time does not exist: the universe contains only space and the physical states within it, changing according to natural laws. Humans use physical processes as references for comparing changes. Within this framework, the stopping of time manifests as absolute stasis of physical states within a specified region. When neighboring regions continue to change, local stasis nevertheless forms part of the same actual history. The past comprises processes that objectively occurred; the present consists of physical states that actually exist; and the future consists of subsequent changes permitted by the current state and natural laws but not yet realized. Historical traces are physical states that still exist in current reality, and restoring an earlier state must likewise be accomplished by physical operations in current reality.

Treating the past as a place that can be arbitrarily visited and rewritten raises two related questions. How can a scene that originally contained no time traveler accommodate someone bearing a future state? And if every alteration generates a new subsequent history, each of which can itself be revisited

and rewritten, what rule determines which history a particular jump connects to? This paper begins from the single-actual-history assumption and distinguishes the course of events that actually occurred from possible subsequent outcomes. A past state can serve as a target for restoration. Once the state within the specified scope is reproduced, the object has returned to the state of the selected moment. At the time of restoration, actual history continues to develop from the then-existing physical state.

This paper adopts a fixed fundamental scale as one of its core hypotheses, consistently denoted by ΔC_0. Relations of state succession, transition intervals, and clock differences each take this parameter as an input. System composition, driving interactions, and environmental conditions enter the specific dynamics, while ΔC_0 remains unchanged when different situations are compared. Modern metrology, relativistic clock differences, statistical physics, and internal-clock studies provide evidence relevant to different propositions of this framework. An additional exponential comparison rule demonstrates the calculation of two-clock readings and shows that, in this example, the observations initially determine only the ratio of the coupling coefficient to the fundamental scale. Deriving the rule from laws of state succession, independently identifying the parameters, and passing tests across different timekeeping mechanisms are necessary steps if the framework is to gain independent explanatory power as a physical theory.

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